

# Differentiable functions – 4

Taylor's formula: remainders and estimates of derivatives

1 сентября 2026 года

## Theoretical minimum.

**Theorem 1 (three forms of the remainder).** For  $f \in C^{n+1}$ ,  $f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x)$ , where the remainder can be written as  $\frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$  (Lagrange),  $\frac{f^{(n+1)}(\xi)}{n!} (x-\xi)^n (x-a)$  (Cauchy) or  $\frac{1}{n!} \int_a^x (x-t)^n f^{(n+1)}(t) dt$  (integral form).

**Theorem 2 (Landau–Kolmogorov).** If  $f$  is twice differentiable on  $\mathbb{R}$  with  $\|f\|_\infty = M_0$  and  $\|f''\|_\infty = M_2$  finite, then  $\|f'\|_\infty \leq \sqrt{2M_0M_2}$ ; on a half-line the constant becomes 2, and both constants are sharp.

**Fact 1.** The working scheme: expand at the point where you know most, and use the remainder as an inequality rather than as an identity. Two Taylor expansions at the endpoints of an interval, added or subtracted, kill the unknown middle terms — this is how all estimates of  $\|f'\|$  through  $\|f\|$  and  $\|f''\|$  are obtained.

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1. Prove all three forms of the remainder and deduce the classical expansions of  $e^x$ ,  $\ln(1+x)$  and  $(1+x)^\alpha$  with explicit bounds for the error on  $[0, \frac{1}{2}]$ .
2. Prove the Landau inequality  $\|f'\|^2 \leq 2\|f\| \|f''\|$  on the line and  $\|f'\|^2 \leq 4\|f\| \|f''\|$  on a half-line, and show that both constants are attained.
3. (Mexmat, 2001) Does there exist a constant  $C$  such that for every continuously differentiable  $f$  on  $\mathbb{R}$ , having a second derivative everywhere except possibly at finitely many points, the conditions  $|f(x)| \leq 1$  and  $|f(x)f'(x)| \leq 1$  at all points where  $f''$  exists imply  $|f''(x)| \leq C$  everywhere?

4. Let  $f \in C^2[0, 1]$  with  $f(0) = f(1) = 0$  and  $\|f''\|_\infty \leq M$ . Prove that  $\|f\|_\infty \leq \frac{M}{8}$  and that the constant  $\frac{1}{8}$  is sharp.
5. (RMO) Let  $\mathcal{C}$  be the set of twice differentiable  $f: [0, 1] \rightarrow \mathbb{R}$  having at least two zeros, counted with multiplicity, and satisfying  $|f''(x)| \leq 1$ . Find  $\max_{f \in \mathcal{C}} \int_0^1 |f(x)| dx$  together with the functions attaining the maximum.